

Analog Communication Laboratory

Experiment 1: Basic Filter Design



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1. Aim

To design, implement, and analyze passive and active filters including Low Pass Filter (LPF), High Pass Filter (HPF), Band Pass Filter (BPF), and voltage Adder Circuit using an operational amplifier.

2. Apparatus

Resistors: 1.5k Ω , 1.8k Ω , 1.2k Ω , 4.8k Ω , 1k Ω

Capacitors: 5.6 nF, 4.7nF, 10nF (by combining capacitors)

Operational Amplifier: UA741 Signal Generator

Digital Storage Oscilloscope (DSO) DC Power Supply

Breadboard and connecting wires

Objectives

1. Task 1:

(a) Design and implement a first-order passive LPF with the specified cut-off frequency of $f_H = 4[(G-1) \pmod{4}] + 12$ kHz. In our case, $G = 16 \Rightarrow f_H = 22$ kHz

(b) Design and implement a first-order passive HPF with the specified cut-off frequency of $f_L = 4[(G-1) \pmod{4}] + 4$ kHz. In our case, $G = 16 \Rightarrow f_L = 16$ kHz

(c) Design and implement a first-order passive and active BPF with above mentioned cutoff frequencies, in our case, $f_H = 22$ kHz and $f_L = 16$ kHz.

i. Find out the 3dB frequency from both time and frequency domain representation of signal on the DSO.

ii. Find out the amplitude of signal at $f_L/2, 2f_H$.

2. Task 2: Design and implement a first-order active LPF using an operational amplifier (Op-Amp) with a specified gain of 2, and observe the output on the DSO.

3. Task 3: Take two voltage signals $x_1(t)$ and $x_2(t)$ and add them using Op-Amp with the formula: $(-1)G(x_1(t) + [(G-1) \pmod{4} + 1]x_2(t))$.

3. Theory

Low Pass Filter (LPF)

A low pass filter allows signals with frequencies lower than a specified cut-off frequency f_H to pass, while attenuating frequencies higher than f_H . It is typically realized using a series resistor and a shunt capacitor. In the frequency domain, the capacitor's reactance X_C decreases as frequency increases, effectively shorting high-frequency signals to ground.

The cut-off frequency, also known as the -3 dB point, is defined as:

$$f_H = \frac{1}{2\pi RC}$$

At this frequency, the output power is half of the input power, and the output voltage is $1/\sqrt{2}$ of the input voltage. The phase ϕ introduced by the LPF is given by :

$$\phi = -\tan^{-1}(2\pi f RC)$$

At $f = f_H$, the phase shift is exactly -45 degrees

The output voltage for the LPF is given by:

$$V_{out} = V_{in} \frac{-jX_C}{R - jX_C}$$

At the cut-off frequency f_H :

$$\frac{-jX_C}{R - jX_C} = \frac{1}{\sqrt{2}}$$

Taking the magnitude and after some algebraic manipulation:

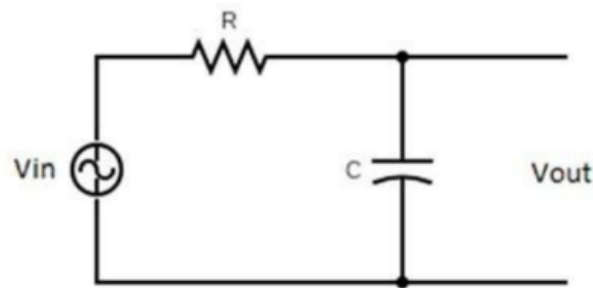
$$\left| \frac{-jX_C}{R - jX_C} \right| = \frac{1}{\sqrt{2}} \Rightarrow \sqrt{\frac{X_C^2}{R^2 + X_C^2}} = \frac{1}{\sqrt{2}} \Rightarrow \frac{X_C^2}{R^2 + X_C^2} = \frac{1}{2} \Rightarrow R^2 = X_C^2 \Rightarrow R = X_C$$

$$R = X_C \Rightarrow R = \frac{1}{C\omega} \quad [\because X_C = \frac{1}{C\omega}]$$

$$\therefore \omega = \frac{1}{RC}$$

The cut-off frequency f_H is:

$$f_H = \frac{1}{2\pi RC}$$



Low Pass Filter Circuit Diagram

High Pass Filter (HPF)

A high pass filter allows signals with frequencies higher than a specified cut-off frequency f_L to pass, while attenuating frequencies lower than f_L . It is realized by placing a capacitor in series with the signal path and a resistor in shunt. At low frequencies, the capacitor acts as an open circuit (high reactance), blocking the signal.

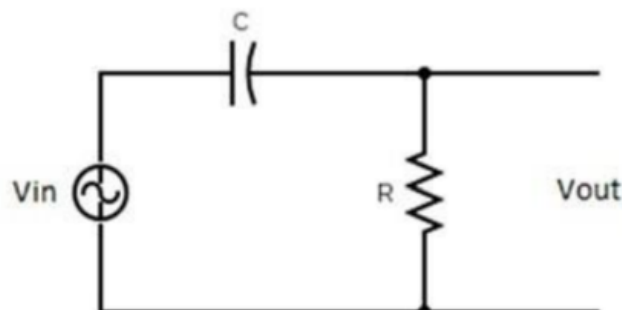
The cut-off frequency is defined as:

$$f_L = \frac{1}{2\pi RC}$$

For frequencies much higher than f_L , the filter passes the signal with minimal attenuation. The phase shift ϕ for a high pass filter is leading and is given by:

$$\phi = \tan^{-1} \left(\frac{1}{2\pi f RC} \right)$$

At the cut-off frequency $f=f_L$, the phase shift is 45 degrees



High Pass Filter Circuit Diagram

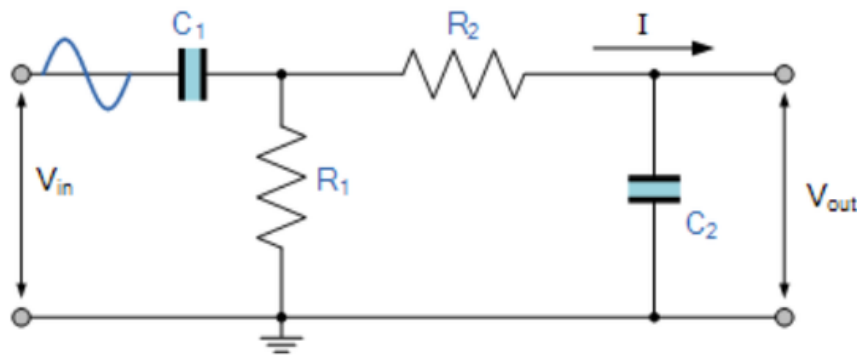
Band Pass Filter (BPF)

A band pass filter allows signals between the lower cut-off frequency f_L and the higher cut-off frequency f_H to pass, while attenuating frequencies outside this range. It is realized by cascading a High Pass Filter (HPF) and a Low Pass Filter (LPF).

The bandwidth (BW) of the filter is given by:

$$\text{Bandwidth} = f_H - f_L$$

The overall gain of the cascaded filter is the product of the individual gains of the HPF and LPF sections. For the frequencies to pass effectively, the filters must be designed such that $f_L < f_H$.



Band Pass Filter formed by cascading HPF and LPF.

Active Band Pass Filter (BPF)

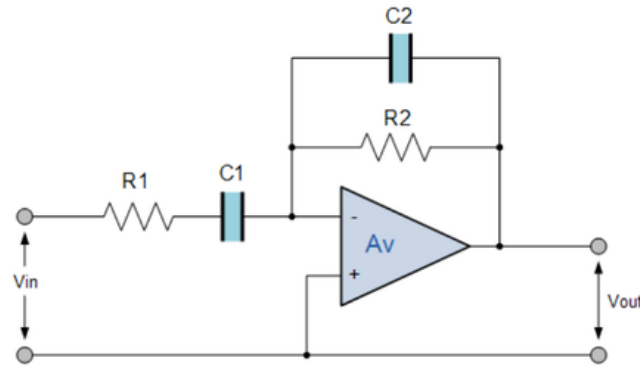
An active Band Pass Filter (BPF) is a frequency-selective circuit that allows a specific range of frequencies to pass while attenuating those below the lower cut-off frequency (f_L) and above the higher cut-off frequency (f_H). A first-order active BPF is typically realized by cascading a first-order high pass filter and a first-order low pass filter with an operational amplifier stage.

The bandwidth (BW) of the filter is defined as:

$$BW = f_H - f_L$$

The passband gain A_v can be adjusted independently of the frequency response via the feedback network of the Op-Amp:

$$A_v = 1 + \frac{R_f}{R_i}$$



Active Band Pass Filter .

Comparison: Active vs. Passive BPF

Active Band Pass Filters offer several technical advantages over their passive counterparts, as summarized in the table below:

Table 1: Comparison between Passive and Active Filters

Feature	Passive Filter	Active Filter
GainCapability	MaxGain ≤ 1 (Lossy)	Can provide amplification (e.g., Gain=2)
LoadingEffect	High (Stages interfere)	Low (Isolated via Op-Amp)
PowerRequirement	None (Self-powered)	Requires DC Supply
FrequencyStability	Poor due to load sensitivity	High stability and sharper response

Active Low Pass Filter

An active low pass filter combines a passive RC network with an operational amplifier (Op-Amp). The Op-Amp provides high input impedance and low output impedance, which prevents the connected load from loading down the filter circuit and altering its characteristics.

The voltage gain (A_v) in the passband for a non-inverting active LPF is determined by the feedback resistors R_f and R_i :

$$A_v = 1 + \frac{R_f}{R_i}$$

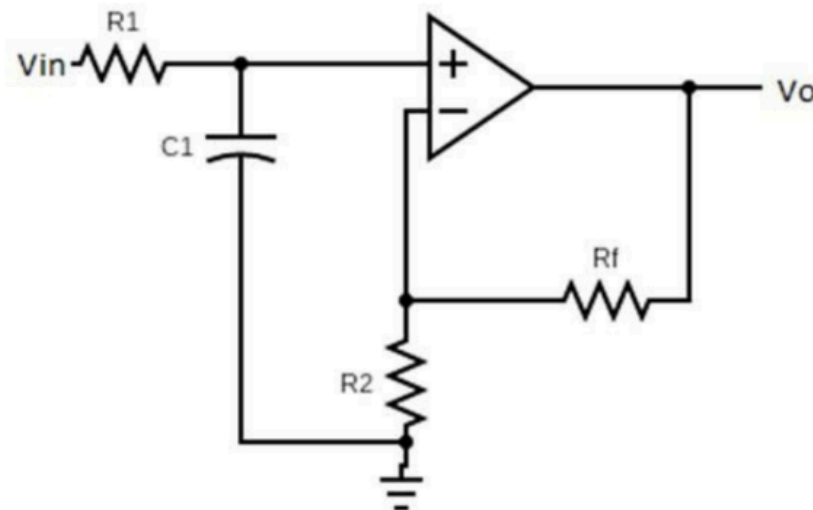
The cut-off frequency f_H is determined by the RC network connected to the non-inverting terminal:

$$f_H = \frac{1}{2\pi RC}$$

The transfer function of the first-order active low pass filter is expressed as:

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{A_v}{1+j(f/f_H)}$$

Key advantages of the active LPF include:



Active Low Band Pass Filter

- **Gain Adjustment:** The passband gain can be easily adjusted using the feedback resistors.
- **Isolation:** The Op-Amp provides excellent isolation between the input and the output.
- **No Insertion Loss:** Unlike passive filters, the output signal strength can be maintained or even amplified.

The frequency response follows a roll-off rate of -20 dB/decade (or -6 dB/octave) for frequencies above f_H .

Voltage Adder using Op-Amp

A voltage adder circuit sums multiple input signals into a single output using an Op- Amp. For the non-inverting voltage adder circuit shown in Figure 5, the output voltage V_{out} is derived by considering the individual contributions of each input source.

The individual output components are given by:

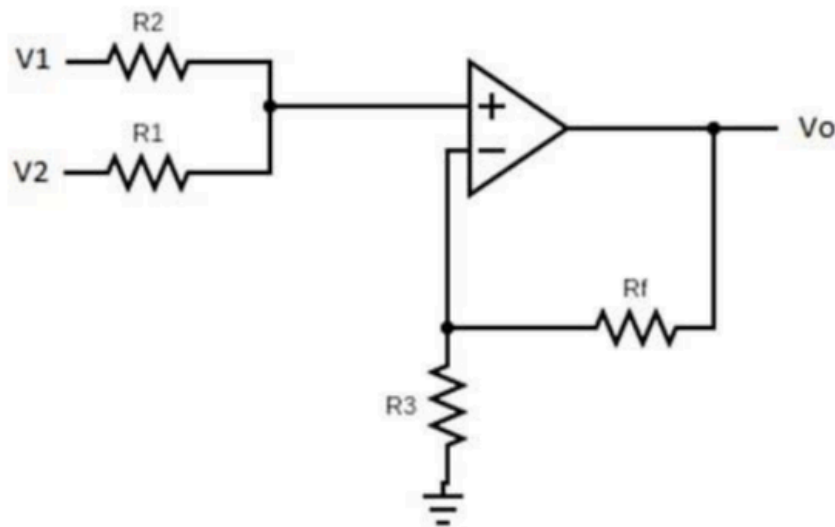
$$V_{O1} = \left(1 + \frac{R_f}{R_3}\right) \times \left(\frac{R_1 V_1}{R_1 + R_2}\right) \quad (1)$$

$$V_{O_2} = \left(1 + \frac{R_f}{R_3}\right) \times \left(\frac{R_2 V_2}{R_1 + R_2}\right) \quad (2)$$

The total output voltage V_{out} is the sum of these components:

$$\therefore V_{out} = V_{O_1} + V_{O_2} \Rightarrow V_{out} = \left(\frac{\frac{V_1}{R_2} + \frac{V_2}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2}}\right) \times \left(1 + \frac{R_f}{R_3}\right)$$

$$\Rightarrow V_{out} = \left(1 + \frac{R_f}{R_3}\right) \left(\frac{R_1 V_1 + R_2 V_2}{R_1 + R_2}\right)$$



Voltage Adder

4. Procedure

1. Circuits were assembled on a breadboard as per the diagrams.
2. Input signal was applied using the signal generator.
3. Output waveform was observed on the DSO.
4. Frequency was varied and corresponding output was recorded.

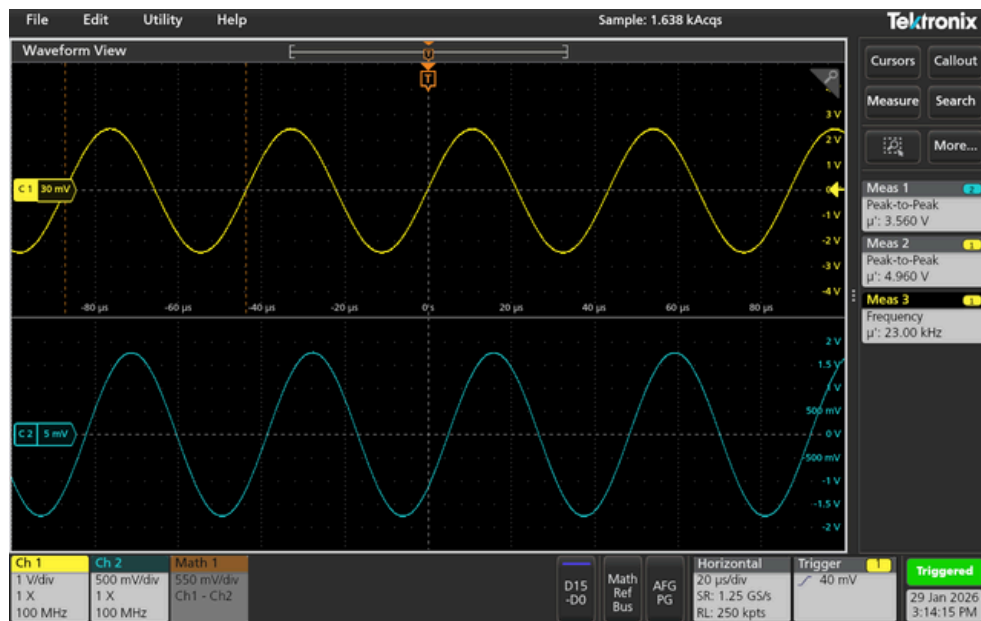
5. Observations and Results

Task 1(a): Low Pass Filter

The values of the components for the LPF were calculated as: $f_H = 14 \text{ kHz}$.

$$\therefore f_H = \frac{1}{2\pi RC} = 22 \text{ kHz} \Rightarrow RC = \frac{1}{2\pi \times 22}$$

Taking $R = 1.5\text{k}\Omega$ we get $C \approx 4.7 \text{ nF}$.

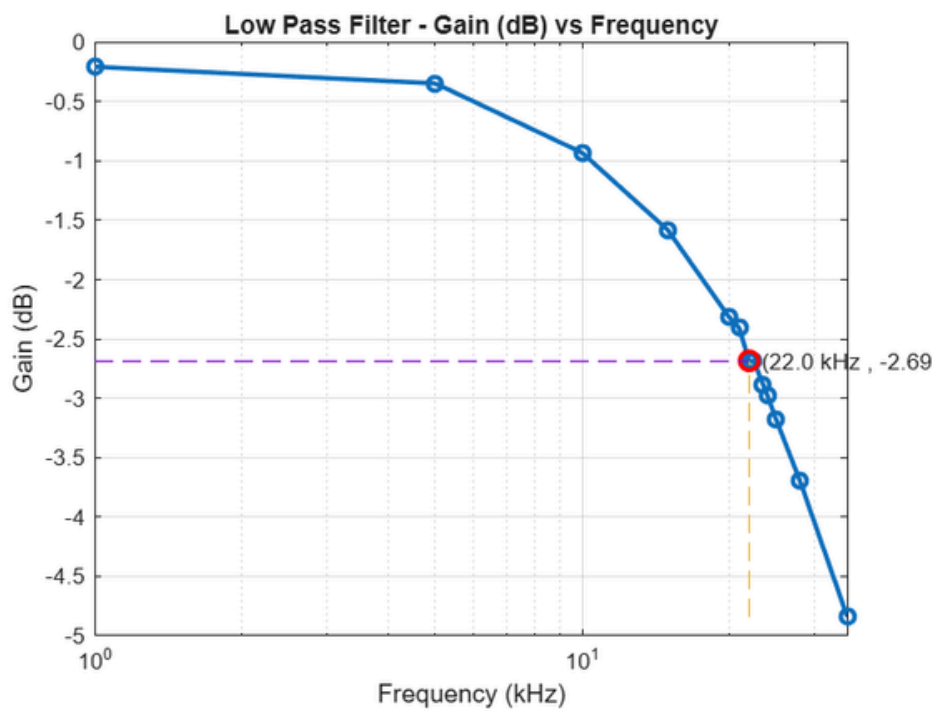


Output at 3dB Point of LPF.

The following table summarizes the measured values for Output Voltage (V_{out}), Input Voltage (V_{in}), Gain, and Phase across varying frequencies as recorded during the experiment.

Table 2: Measured Frequency Response Data for LPF

Frequency (KHz)	$V_{out}(V)$	$V_{in}(v)$	23.5	3.56	4.96
1	5	5.12	24	3.52	4.96
5	4.92	5.12	25	3.44	4.96
10	4.6	5.12	28	3.24	4.96
15	4.2	5.04	35	2.84	4.96
20	3.8	4.96			
21	3.76	4.96			
22	3.64	4.96			
22.5	3.64	4.96			



Frequency Response of LPF

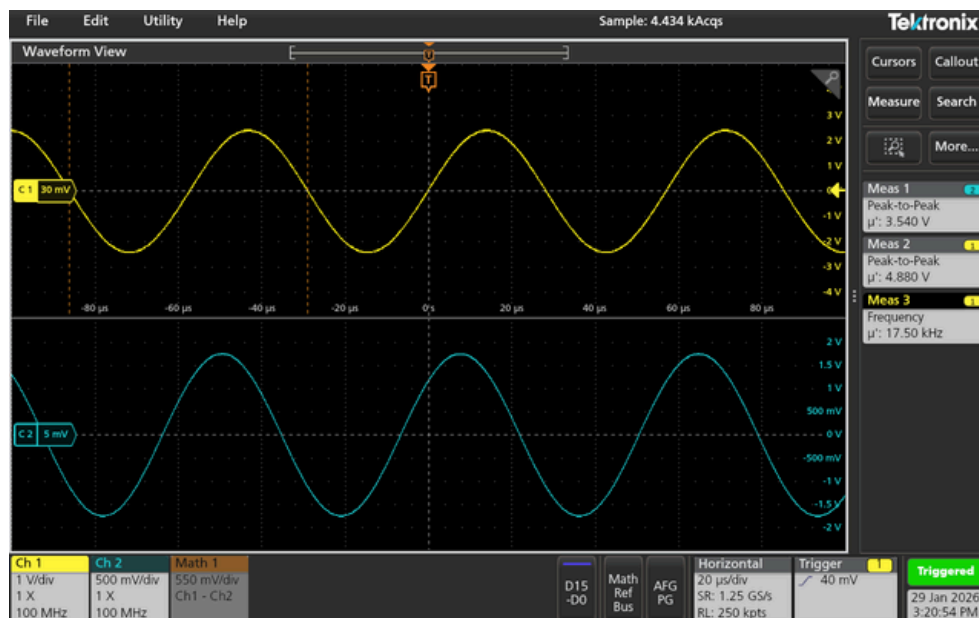
The 3dB cutoff frequency of the LPF is approximately 22 kHz. This is the frequency at which the gain of the filter is 3 dB below its maximum gain.

Task 1(b): High Pass Filter

The values of the components for the HPF were calculated as: $f_L = 16 \text{ kHz}$.

$$\therefore f_L = \frac{1}{2\pi RC} = 16 \text{ kHz} \Rightarrow RC = \frac{1}{2\pi \times 16}$$

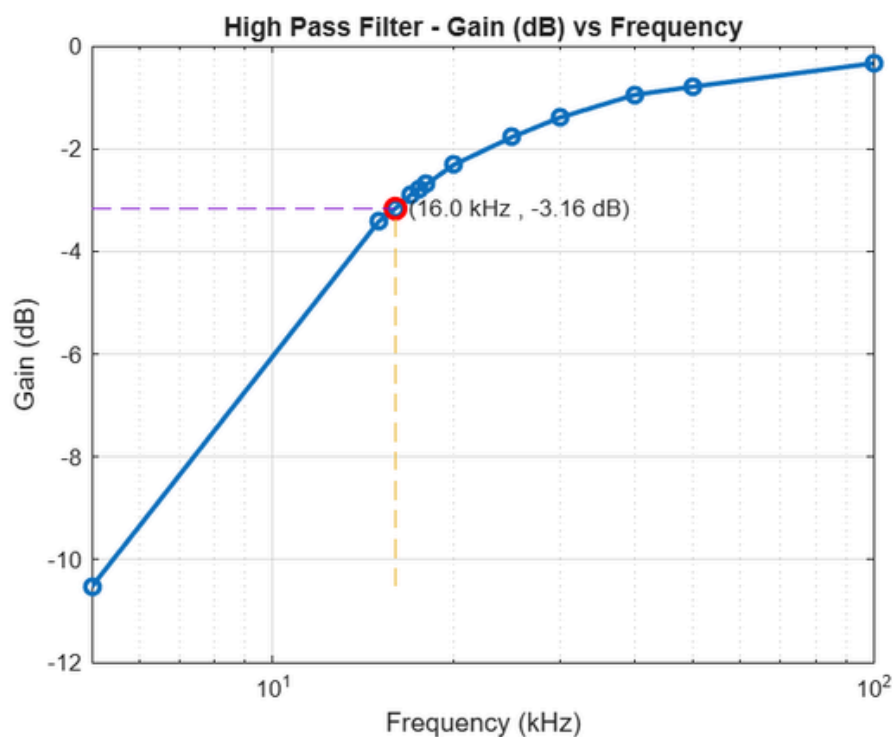
Taking $R = 1.77 \text{ k}\Omega$ we get $C \approx 5.6 \text{ nF}$.



Output at 3dB Point of HPF.

The input voltage was maintained at $V_{in} = 5 \text{ V}$ throughout the frequency sweep.

Frequency (KHz)	Vout(V)	Vin(V)	20	3.74	4.88
			25	3.98	4.88
5	1.5	5.04	30	4.14	4.86
15	3.32	4.92	40	4.34	4.84
16	3.42	4.92	50	4.42	4.84
17	3.5	4.88	100	4.62	4.8
17.5	3.54	4.88			
18	3.58	4.88			



Frequency response of the High Pass Filter.

Task 1(c): Band Pass Filter

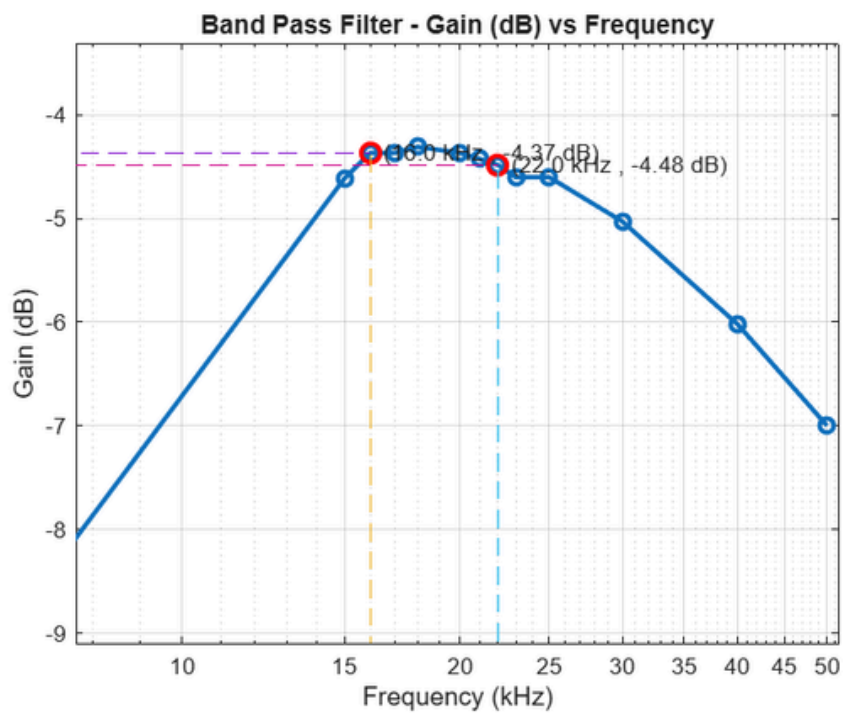
For the Band Pass Filter, the frequency is given by $f = \frac{1}{2\pi RC}$.

- $f_H = 22$ kHz, taking $C = 4.7$ nF and $R = 1$ k Ω .
- $f_L = 16$ kHz, taking $C = 5.6$ nF and $R = 1.77$ k Ω .

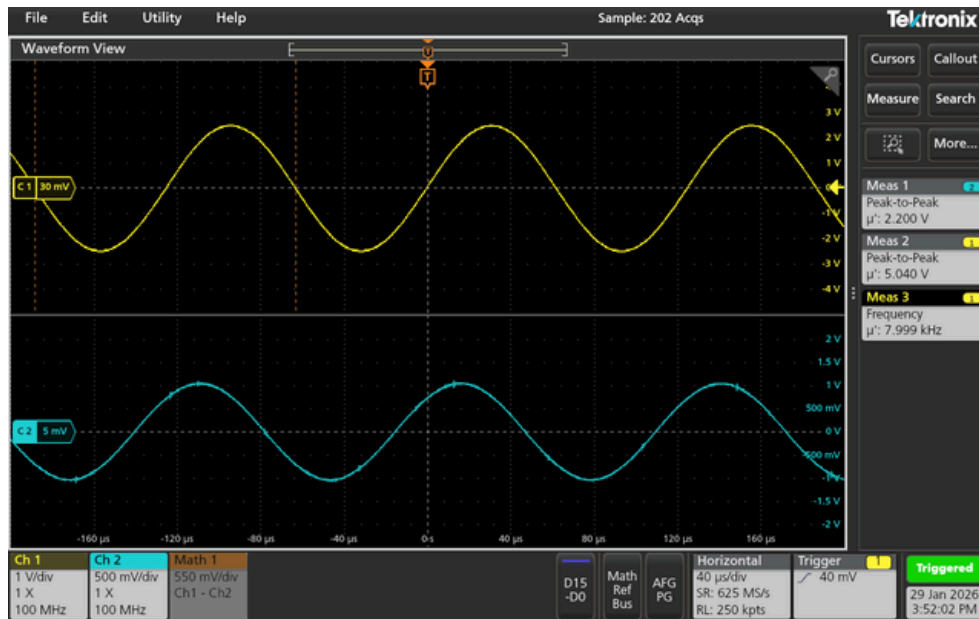
Table 4: Measured Frequency Response of the Band Pass Filter

Frequency (KHz)	Vout(V)	Vin(V)
5	1.54	5.04
15	2.94	5
16	3	4.96
17	3	4.96
18	3.02	4.96
20	3	4.96
21	2.98	4.96
22	2.96	4.96

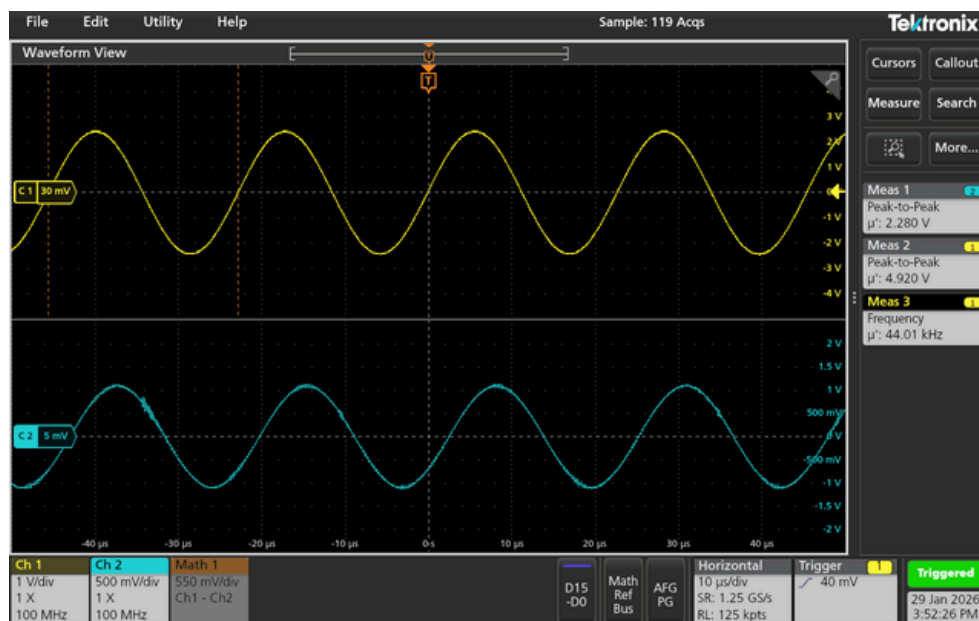
23	2.92	4.96
25	2.92	4.96
30	2.78	4.96
40	2.4	4.8
50	2.18	4.88



Frequency response of the Band Pass Filter .



Wave at $f_L/2$ of band pass filter



Wave at $2f_H$ of band pass filter

Task 2: Active Low Pass Filter

We implemented the active LPF using the Op-Amp UA741. The passband gain is set as:

$$\text{Gain} = \left(1 + \frac{R_f}{R_2}\right) = 2 \Rightarrow R_f = R_2$$

The target cut-off frequency is:

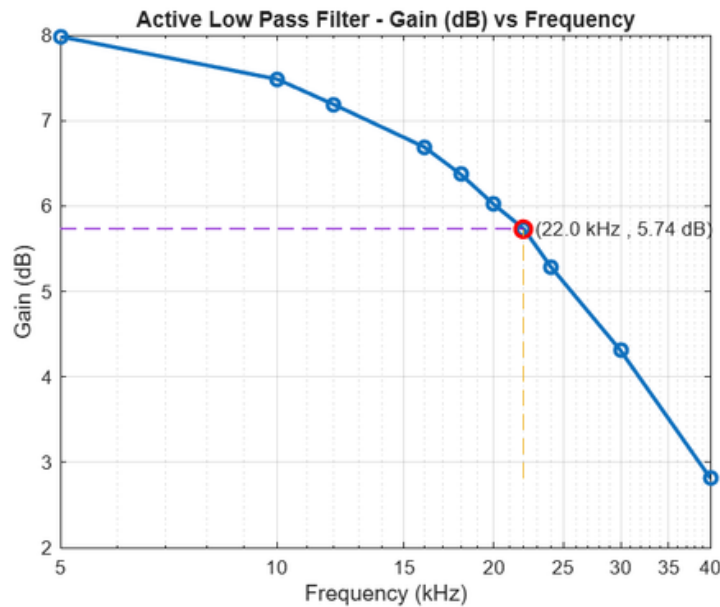
$$f_H = \frac{1}{2\pi RC} = 22 \text{ kHz}$$



Figure 13: Output at 3dB Point of Active LPF .

Table 6: Measured Frequency Response Data for Active LPF

Frequency (KHz)	Vout(V)	Vin(V)
5	12.64	5.04
10	11.84	5
12	11.44	5
16	10.72	4.96
18	10.33	4.96
20	9.92	4.96
22	9.6	4.96
24	9.12	4.96
30	8.08	4.92
40	6.8	4.92



The 3dB cutoff frequency of the Active LPF is approximately 24 kHz, where the gain is 3 dB below its maximum value.

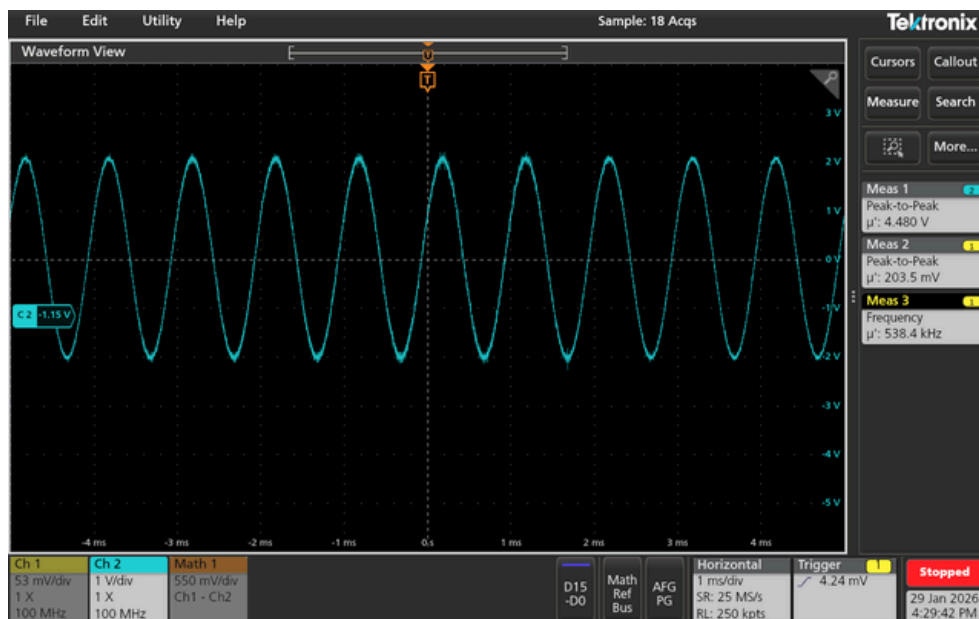
Task 3: Voltage Adder Circuit using Op-Amp

We have used the Op-Amp UA741 for our circuit. The output voltage V_{out} is given by:

$$V_{O1} = \left(1 + \frac{R_f}{R_3}\right) \times \left(\frac{R_1 V_1}{R_1 + R_2}\right), V_{O2} = \left(1 + \frac{R_f}{R_3}\right) \times \left(\frac{R_2 V_2}{R_1 + R_2}\right)$$

$$\therefore V_{out} = V_{O1} + V_{O2} \Rightarrow V_{out} = \left(1 + \frac{R_f}{R_3}\right) \left(\frac{R_1 V_1 + R_2 V_2}{R_1 + R_2}\right)$$

For our specific case, we required $y_{out} = x_1(t) + 4 \cdot x_2(t)$. By taking $R_1 = R_3 = 1.2k\Omega$ and $R_2 = R_f = 4.8k\Omega$, we achieve it.



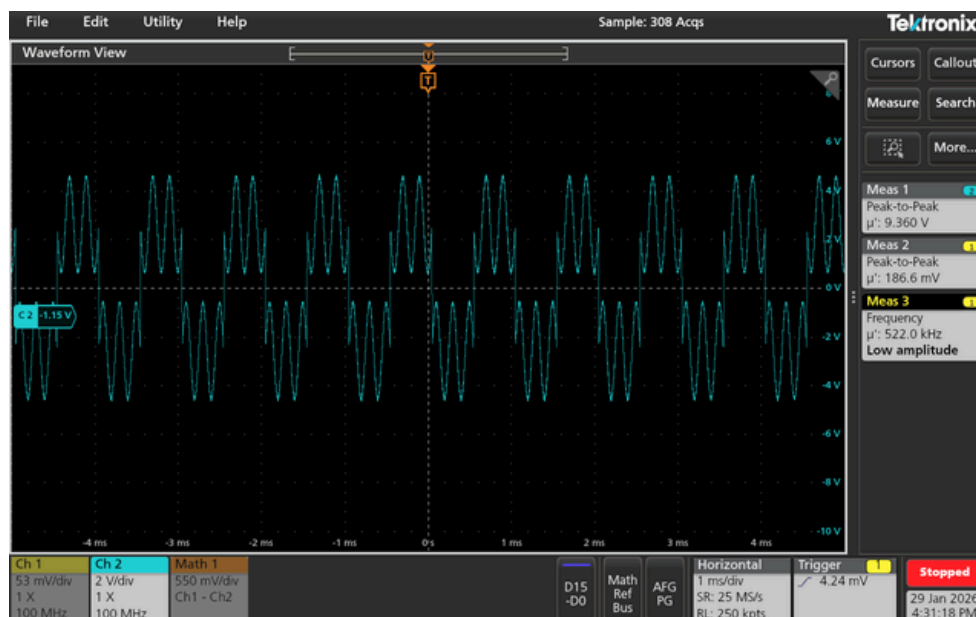
Voltage Adder Output for $V_1(t) = \sin(2\pi 1000t)$ and $V_2(t) = 1 \sin(2\pi 1000t)$.



Voltage Adder Output for $V_1(t) = 5\sin(2\pi \cdot 1000t)$ and $V_2(t) = \sin(2\pi \cdot 5.1000t)$.



Voltage Adder Output for $V_1(t) = 10\sin(2\pi \cdot 5000t)$ $V_2(t) = \text{Pulse } 10\text{Khz } 5\text{Vpp}$



Voltage Adder Output for $V_1(t) = \text{Square } 5\text{Vpp}$ $V_2(t) = \sin(2\pi \cdot 5000t)$

6. Discussion

Passive Filters (Low Pass, High Pass and Band Pass)

In this experiment, passive first-order RC filters were designed and implemented using resistors and capacitors. Since passive filters do not contain active elements, they are unable to provide amplification and only shape the frequency response of the signal.

For the **Low Pass Filter (LPF)**, the theoretical cutoff frequency was designed to be approximately 22 kHz. From the experimental observations on the DSO, the cutoff frequency was found to be close to the theoretical value. In the low-frequency region, the gain remained nearly constant, and as the frequency approached cutoff, the gain reduced gradually at a slope of approximately -20 dB/decade, as expected for a first-order filter. The phase shift approached -45° near the cutoff frequency.

For the **High Pass Filter (HPF)**, the designed cutoff frequency was approximately 16 kHz. Experimentally, the cutoff frequency was observed close to this value. At low frequencies, the gain was very small, and as frequency increased beyond cutoff, the gain approached nearly 0 dB. The slope near cutoff followed the expected $+20$ dB/decade, and the phase shift approached $+45^\circ$ near the cutoff frequency.

The **Passive Band Pass Filter (BPF)** was implemented by cascading the HPF and LPF sections, with lower cutoff at 16 kHz and upper cutoff at 22 kHz. The passband was observed between these frequencies; however, the gain in the passband was significantly less than unity.

This reduction in gain occurred mainly due to:

- Loading effects between stages
- Impedance interaction
- Additional voltage division between cascaded sections

Since the output impedance of the HPF became comparable to the input impedance of the LPF, the second stage drew current from the first stage. This altered the effective RC time constants and slightly shifted the cutoff frequencies. Because the separation between lower and upper cutoff frequencies was relatively small, the -3 dB points did not perfectly coincide with the theoretical values. Thus, although the filter shape was correct, attenuation was higher than the ideal theoretical case.

Active Low Pass Filter

To overcome the limitations of passive filters, a first-order active low pass filter was implemented using a UA741 op-amp in non-inverting configuration. The designed passband gain was:

$$A_v = 1 + R_f/R = 16$$

Experimentally, the gain in the passband was close to the designed value (≈ 24 dB), and the cutoff frequency was observed near 22 kHz, which closely matched the theoretical calculation.

Compared to the passive LPF:

- The roll-off near cutoff was more pronounced.
- The passband gain was amplified (unlike passive filters).
- The response was more stable.

Low values of feedback resistors were selected to minimize the effect of parasitic capacitances across the feedback network, allowing the circuit to operate effectively over a wider frequency range.

Minor deviations from theoretical values occurred due to:

- Finite bandwidth of the op-amp
- Input offset voltage
- Parasitic capacitance
- Probe loading effects

Active Band Pass Filter and Buffering Effect

An active band pass filter was implemented to reduce loading effects observed in the passive BPF. The op-amp provided:

- Very high input impedance
- Low output impedance
- Isolation between stages

Because of this buffering action, impedance interaction between stages was minimized. The passband gain improved significantly compared to the passive bandpass filter, and cutoff frequencies were more stable.

Although attenuation still occurred before the signal could fully reach V_{in} after exceeding the lower cutoff frequency, the overall performance was much closer to theoretical expectations than in the passive case.

The op-amp acted as an effective buffer, preventing loading from propagating through the filter chain.

Op-Amp Power Supply Considerations

The op-amp was powered using ± 12 V supply rails. Since the LM741 is not a rail-to-rail op-amp and the signal amplitudes were in the range of 0–5 V, ± 12 V provided sufficient headroom for proper operation without clipping. However, this choice increases overall power consumption.

Voltage Adder Using Op-Amp

The voltage adder was implemented using an op-amp in feedback configuration to verify linearity and superposition properties. By selecting appropriate resistor values, the circuit successfully produced weighted summation of input signals.

In addition to basic sinusoidal inputs, we also tested:

- Square wave summation
- Ramp signal summation
- Combination of different waveform types

The output observed on the DSO confirmed:

- Proper superposition of signals
- Linear operation of the op-amp
- Output amplitude closely matching theoretical weighted sums

The adder demonstrated the fundamental linear property:

$$V_{out}=A_1V_1+A_2V_2+...V_{out} = A_1V_1 + A_2V_2 + ...$$

This verified the accuracy of op-amp based summation circuits.

Overall Observations

- Passive filters showed expected frequency shaping but suffered from attenuation and loading effects.
- Active filters provided amplification, better stability, and isolation.
- Experimental values were close to theoretical predictions.
- Minor deviations were due to practical non-idealities such as component tolerances, parasitic effects, and non-ideal op-amp behavior.
- The experiment also improved practical understanding of breadboard connections, DSO measurements, signal generators, and op-amp configurations.